

DAVID SMITH

Computer Science & Economics Student

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🌐 linkedin.com/in/davidchrsmith

🌐 davidchrsmith.github.io/PortfolioWebsite

Education

University of Minnesota Twin-Cities

Expected May 2027 (Senior)

Dual Degree: B.S. in Computer Science and B.S. in Economics (GPA: 3.544)

Minneapolis, MN

- **Relevant Coursework:** Advanced Algorithms and Data Structures, Software Engineering I, Fundamentals of Machine Learning, Database Systems, Dev. of Secure Software Systems, Industrial Organization & Antitrust Policy, Environmental Economics
- CSE Dean's List, Spring 2024, Spring 2025, and Spring 2026

Projects

Personal Portfolio Website

- Designed and developed a responsive personal portfolio website using **React**, **JavaScript**, **CSS**, to showcase my technical projects and skills.
- Created an interactive user experience featuring modern UI design, smooth animations, and 3D visual elements using **Three.js**.
- Integrated **EmailJS** to provide a functional contact form, allowing visitors to send messages directly through the website without requiring a custom backend.

Drone Package Delivery Simulation

- Implemented a drone delivery simulator in **C++** with Docker containerization, optional theft events, and a notification system for completed deliveries.
- Developed multiple path-finding algorithms for route optimization and evaluated efficiency across strategies.
- Demonstrated skills in algorithm design, object-oriented programming, and simulation of real-world systems.

Goods4Goods App (In Progress)

- Building a cross-platform mobile and web app using **React Native** and **PostgreSQL** enabling users to trade items without cash transactions.
- Designed and implemented key user interfaces and image upload features for streamlined usability.

Experience

HaFSBX

May 2026 – Present

Research Intern

Maplewood, MN

- Developed full-stack web applications using **React**, **Next.js**, **Django REST Framework**, **CSS**, and **Three.js**, implementing new features and improving user experience.
- Designed, implemented, tested, and deployed ROS2 robotic applications using **Python**, **C++**, and XML launch/configuration files.
- Developed and tested REST APIs and backend data pipelines using **OpenAPI** and **Bruno** for API design, documentation, and validation.
- Built and validated robotic workflows using **NVIDIA Isaac Sim**, Mission Control, and Mission Dispatch for simulation and deployment.

Lavner Education

June 2025 – Aug. 2025

Internship - STEM Instructor

St. Paul, MN

- Taught and mentored students (ages 6–14) in Python, C++, robotics, and AI with Ethics, fostering problem-solving and teamwork skills.
- Collaborated with instructors to refine curriculum, adding interactive coding challenges and hands-on projects that improved student engagement and project completion rates.
- Debugged student code and robotics programs in real time, providing individualized guidance and reinforcing core programming concepts.
- Troubleshoot technical issues with laptops, IDEs, and robotics kits, minimizing class disruptions and ensuring smooth delivery of lessons.

Technical Skills

Languages: Python, C++, Java, JavaScript, SQL, OCaml

Frameworks: React, Next.js, React Native, Django, Django Rest Framework, Three.js

Robotics: ROS2, ROS, Rviz, IntelliJ

Tools: Git, GitHub, Docker, Linux, Bruno, Figma, RStudio